SAR-RAG — Research Status After Screening v3

Training-only retrieval screening, n = 286 examples | 9,724 result rows | **zero LLM calls** | no evaluation split accessed | 14 August 2026

How to read this report

Label	Meaning
RAW	Measurement reproduced directly from the per-example CSV
STAT	Supported by the pre-registered test and multiplicity correction
INFER	Reviewer interpretation beyond the test result
LIMITATION	A constraint on what can be concluded
BLOCKED	Not executed; explicitly gated

1. Decision

Question	Answer
Is representation routing scientifically justified?	**No.** Fails the frozen rule on all three limbs.
Does Oracle@2 show pair complementarity?	**No — exactly zero.** A pair never beat the best single view on any of 210 questions.
Does the router approach the oracle ceiling?	Yes: router 0.7376 vs Oracle@2 0.7575. The ceiling itself is only +0.0249 over a constant policy.
Is a constant single-view policy stronger?	Statistically indistinguishable (semantic 0.7326 vs router 0.7376).
Retain hierarchical_v2?	As a code correctness fix yes; as a rescue for the routing claim no.
Do novelty / gap / utility / redundancy affect outputs?	No. Novelty and gap change 0 of 286 examples; utility and redundancy bind 0 times.
Budget-matched advantage over Traditional RAG?	**No.** The matched-budget comparison is null; unmatched favours the baseline.

[DECISION] **FINAL: PIVOT TO ADAPTIVE SINGLE-VIEW RETRIEVAL.** This is a decisive negative result, not an inconclusive one. The attainable ceiling over all four views and all six pairs sits +0.025 above always using the semantic view; the frozen protocol required +0.05. Repairing the weakest representation by +0.247 recall moved that ceiling by exactly 0.000.

2. Artifact validation

Check	Expected	Observed	Verdict
Result rows	286 x 34 = 9,724	9,724	PASS
Unique examples	286	286	PASS
Conditions	34, each n=286	34, each with 286 rows	PASS
Duplicate keys	0	0	PASS
Partition sizes	194 tuning / 92 selection	194 / 92	PASS
Calibration-ID overlap	0	0	PASS
Prior-diagnostic-ID overlap	0	0	PASS
Evaluation-split IDs present	0	0	PASS
Frozen manifest SHA-256	5c570ec2b4c52e1b...a54d05	identical; git clean	PASS
LLM / Groq access	none	RetrievalOnlyAnswerer only	PASS

[RAW] **NaN audit.** 2,584 NaN cells per metric is not corruption: 2,584 = 76 x 34, and those 76 examples are exactly the ones with zero gold evidence spans, where recall is undefined. Verified cell-by-cell — a metric is NaN if and only if gold_spans = 0. Effective scored sample: **210 of 286**.

Partition	QASPER	MuSiQue	Total scored	Use
selection	22	45	**67**	**design decisions**
tuning	48	95	143	descriptive only
pooled	70	140	210	descriptive only

[LIMITATION] The QASPER selection cell (n = 22) and the QASPER INSUFFICIENT-long stratum (11 total, 4 in selection) cannot support standalone claims. Per-dataset selection results are directional only.

3. Primary comparison — selection partition, evidence recall

Reference is the frozen full SAR-RAG (mean recall 0.7226). n = 67 paired. Holm family = 33 conditions within (metric, partition, scope).

Condition	Mean	Diff	95% CI	p	Holm p
sliding_window_k20	0.8085	+0.0858	[+0.036, +0.141]	0.0007	0.0165
sliding_window_matched_k20	0.7774	+0.0547	[+0.007, +0.109]	0.0355	0.6762
full_hierarchical_v2	0.7376	+0.0149	[+0.000, +0.045]	1.0000	1.0000
semantic_only	0.7326	+0.0100	[-0.015, +0.045]	1.0000	1.0000
hierarchical_v2_only	0.7189	-0.0037	[-0.049, +0.041]	1.0000	1.0000
select_one_representation	0.7152	-0.0075	[-0.022, +0.000]	1.0000	1.0000
paragraph_only	0.7127	-0.0100	[-0.045, +0.015]	1.0000	1.0000
no_novelty	0.7226	+0.0000	[+0.000, +0.000]	1.0000	1.0000
no_similarity_gap	0.7226	+0.0000	[+0.000, +0.000]	1.0000	1.0000
no_post_retrieval_deduplication	0.7226	+0.0000	[+0.000, +0.000]	1.0000	1.0000
sliding_window_k10	0.6915	-0.0311	[-0.070, -0.004]	0.1249	1.0000
sentence_window_only	0.6629	-0.0597	[-0.114, -0.005]	0.0349	0.6762
hierarchical_only	0.4726	-0.2500	[-0.336, -0.169]	0.0000	0.0016
rrf_probe_only_no_adaptive	0.4092	-0.3134	[-0.402, -0.231]	0.0000	0.0016

[STAT] Exactly one condition beats the frozen primary after Holm correction: **sliding_window_k20 (+0.0858, Holm p = 0.0165)**.
[STAT] Adaptive retrieval past the top-3 probe is the one component with a large, unambiguous, corrected effect (rrf_probe_only_no_adaptive –0.3134, Holm p = 0.0016).

4. The shared budget is nominal, not binding

System	Context tokens	% of 2,000 cap	Final k	Recall
full (SAR-RAG)	1,305.2	**65%**	15.22	0.7226
sliding_window_k20	1,857.7	**93%**	16.52	0.8085
sliding_window_matched_k20	1,780.2	89%	19.30	0.7774
sliding_window_k10	1,244.1	62%	9.83	0.6915

[RAW] SAR-RAG stop reasons (selection, n = 92): all representations paused 63.0%, final-chunk safety cap 30.4%, retrieval-round cap 6.5%. **The token budget never appears as a stop reason**; only 12 of 92 runs reach 95% of the cap. **[INFER]** SAR-RAG is halted by patience and the 20-chunk cap while ~700 tokens of its allowance sit unused, whereas the baseline spends 93% of the same allowance. sliding_window_k20 therefore wins +0.0858 recall while consuming **42.3% more context**. At comparable spend the advantage vanishes in both directions: sliding_window_k10 is –0.031 (p = 0.125) and matched_k20 does not survive Holm (p = 0.676). **Every Traditional comparison in this project is confounded until the arms are matched on tokens actually consumed, not on chunk count.**

Stability of the k20 result	Diff	p	Holm p
selection (n = 67, decision basis)	+0.0858	0.0007	**0.0165**
tuning (n = 143, descriptive)	+0.0335	0.235	1.000
pooled (n = 210, descriptive)	+0.0502	0.017	0.304
MuSiQue only (selection, n = 45)	+0.1352	0.0001	**0.0033**
QASPER only (selection, n = 22)	–0.0152	1.000	1.000

[LIMITATION] *The magnitude is not stable across partitions and is carried entirely by MuSiQue. It is a serious unresolved threat to the baseline comparison, not a finished finding.*

5. Oracle@1 and Oracle@2 — pair complementarity is exactly zero

Scope	n	Oracle@1	Oracle@2	O2 – O1	Best constant	Router	O2 – const
selection	67	0.7575	0.7575	**0.0000**	0.7326	0.7376	+0.0249
sel · QASPER	22	0.5758	0.5758	**0.0000**	0.5758	0.5758	+0.0000
sel · MuSiQue	45	0.8463	0.8463	**0.0000**	0.8111	0.8167	+0.0352
tuning	143	0.8415	0.8415	**0.0000**	0.8065	0.7976	+0.0350
pooled	210	0.8147	0.8147	**0.0000**	0.7766	0.7785	+0.0381

[RAW] Per-example verification over all 210 scored questions and all six forced pairs: a pair strictly beats the best single view on **0 of 210** examples in the candidate system and 1 of 210 in the legacy system; a pair is the unique maximiser on **0 of 210**. [STAT] Oracle@2 exceeds the best constant view by +0.0249, CI [0.000, +0.060], p = 0.251 — below the frozen 0.05 margin and not significant. [INFER] Under a shared budget a second representation splits the budget and never recovers evidence the best single view misses. The "one or two complementary representations" premise has no headroom to exploit — this is not merely a router that fails to exploit it.

[LIMITATION] 65 of 67 selection examples have two or more tied maximisers, so which view is best remains unreliable and no per-view attribution is made. The oracle value is a maximum and is unaffected by ties.

6. Hierarchical v2 — representation fixed, system unmoved

Comparison	Diff	95% CI	p	dz
hierarchical_v2_only – hierarchical_only (selection, n=67)	**+0.2463**	[+0.153, +0.341]	0.00005	+0.63
same, tuning (n=143)	+0.2476	[+0.174, +0.321]	0.00005	+0.55
same, pooled (n=210)	+0.2472	[+0.189, +0.305]	0.00005	+0.58

Frozen retention requirement	Threshold	Observed	Met
v2 improves over v1	> 0	+0.2463	YES
candidate primary noninferior to legacy	>= –0.02	+0.0149	YES
AND recall gain	>= +0.02	+0.0149	NO
OR context reduction	>= 10%	+0.2%	NO
OR Oracle@2 gain over legacy set	>= +0.02	0.0000	NO
RETAINED			**NO**

[INFER] This is the most informative result in the experiment. Repairing the worst representation by +0.246 recall moves the full system by +0.015 and moves the attainable ceiling by exactly 0.000. A pipeline whose ceiling is invariant to a quarter-point improvement in one of its four views does not derive its value from representation selection. hierarchical_v2 should still replace v1 as a correctness fix, with v1 preserved for the record; what the rule denies is the *claim* that the repair rescues routing.

7. Gate components — what actually changes an output

Component	Examples with ANY output change	Fields changed	Recall diff	Verdict
Novelty rule (0.2292)	**0 / 286**	none	0.0000	**INERT**
Similarity-gap rule (1.9052)	**0 / 286**	none	0.0000	**INERT**
Post-retrieval deduplication	56 / 286	final_k, context_tokens, precision	0.0000	Efficiency only
Second representation	48 / 286	recall on only 4, CER on 3	–0.0075	Negligible

Internal instrumentation, n = 286 (6,989 candidates scored, 2,000 batches). max_* fields are aggregated with max, not summed.

Gate	Reached	Would reject	Binding
Relevance percentile	6,989	1,788	**1,788**
Novelty (0.2292)	5,201	108	107
Marginal utility (0.1289)	—	135	**0**
Redundancy (0.9547)	—	—	**0**
Budget caps	—	—	478

Similarity-gap rule	Value
Eligible batches (rule may fire)	664
Times the rule fired	**0**
Maximum standardized gap ever observed	**1.0925**
Maximum gap on an eligible empty batch	1.0119
Frozen threshold	**1.9052**

[STAT] The similarity-gap rule is not merely unused — it is **unreachable**. The largest standardized gap anywhere in 286 examples is 1.0925, 43% below the frozen threshold of 1.9052. **[STAT]** Marginal utility and redundancy bind **0 times in 6,989 candidate evaluations**. **[INFER]** Novelty binds 107 times yet changes no output, meaning every candidate it rejected would have been removed downstream anyway by budget caps or deduplication.

8. Failure analysis

Class	Count	Note
Complete evidence recovered	129	
Partial	55	
Total miss (recall = 0)	26	**18 of these are unreachable by any view or pair**
Recoverable by perfect routing	**14 / 210**	3 of 67 in the selection partition

[INFER] The dominant failure mode is evidence that is not retrievable by any configuration of this embedding and chunking family — an encoder/corpus problem, not a routing problem. Routing repair addresses at most 6.7% of scored examples.

9. What survives, what is dropped, what blocks the paid run

Survives into v2	Dropped from the claim
Adaptive retrieval past a shallow probe (–0.313 without it, Holm 0.0016) A single representation (semantic; paragraph and hierarchical_v2 indistinguishable) Post-retrieval deduplication as an explicit efficiency mechanism Relevance-percentile scoring — the only gate that binds	RRF representation selection The one-or-two-view premise The novelty rule The standardized similarity-gap rule Marginal utility and redundancy as components Any assertion of superiority over Traditional RAG

#	Blocking prerequisite before any paid run
1	**Fix the budget defect first.** SAR-RAG spends 65% of its stated token budget while the baseline spends 93%. Re-screen free, matched on tokens consumed.
2	**Recalibration is mandatory.** The frozen thresholds were fitted with the four-view router and hierarchical v1 in the pipeline. A single-view method is a different estimand; the v1 manifest cannot carry over.
3	Resolve the sliding_window_k20 instability (selection +0.086 / tuning +0.034) on training data.
4	Verified current Groq rates on the run date, and explicit owner authorisation.

[BLOCKED] **The paid held-out evaluation remains blocked.** Nothing in this analysis authorises it. Running it against the current method would spend the budget confirming a contribution this experiment has just refuted.